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Project Title

FocusFeed

Project Summary

We are building a personalized recommendation platform designed to unify and streamline the consumption of current events and news. By aggregating content from multiple sources including news outlets, social media platforms, and online forums, FocusFeed creates a comprehensive, tailored feed for each user. The platform uses advanced recommendation algorithms to curate content based on user interests and interactions, providing a seamless and engaging news consumption experience.

Users begin by selecting their interests across various categories such as technology, climate, sports, or global events. The platform then leverages its recommendation algorithm to build a dynamic feed of article compilations tailored to these preferences. Through interactive features like swipe-based navigation and a like/dislike system, users can further refine their content preferences, allowing the platform to continuously improve its recommendations.

Description

In today's digital landscape, news and information are highly fragmented across numerous platforms and sources. Users seeking comprehensive information about specific topics must navigate through multiple websites, social media platforms, blogs, and news outlets. This fragmentation creates several challenges, some of which are, time-consuming information gathering process, inconsistent quality and reliability of sources, difficulty in tracking related discussions and perspectives, information overload and an overwhelming user experience.

FocusFeed addresses these challenges by aggregating content from multiple trusted sources into a single, cohesive feed, implementing intelligent content curation based on user preferences, providing real-time updates and discussions from various platforms, and creating a streamlined, intuitive user interface for content discovery. We are building an all in one platform for people to learn about current events and learn what others think about these as well.

Creative Components

1. Interactive Topic Network Visualization

Using D3.js and Cytoscape.js libraries to create a complex, interactive visualization:

- Dynamic force-directed graph showing topic relationships
- Real-time updates as new content is aggregated
- Custom physics-based animations for node interactions
- Zoom and pan capabilities with detail-level adaptation
- Interactive filtering and highlighting based on topic relationships
- Custom WebGL rendering for handling large numbers of nodes efficiently
- If there is enough time we can build out HTTP polling for this for quicker updates

Summary: Every person will have access to an interactive map of article feeds and where they are located on a map, based on a custom implemented map that we will build from scratch using some basic libraries for the base level design. People can filter based on what they want to see in the map, and we will be able to efficiently make users query's happen in real time. For example, say someone is really interested in sports, they can filter the map down to just sports and the map would show a network of sport "feed notes." For definition sake a feed note is our combination of articles, tweets, reddits, comments etc… We get all this data from the real time apis, so the user could click on sport and see a giant graph of all interconnected nodes that represent feed notes for different sports across the world.

2. Multi-Source Real-Time Integration System

A sophisticated system that intelligently combines and processes content from multiple sources (news APIs, social media, forums) in real-time to create unified, meaningful feed notes:

- Custom middleware architecture that:
  - Handles parallel requests to 5+ different APIs efficiently
  - Implements smart rate limiting to prevent API throttling
  - Manages complex error scenarios and API downtime
  - Optimizes request patterns based on API response times
- Real-time content processing:
  - Natural Language Processing for content analysis
  - Semantic similarity detection for related content
  - Custom caching system for frequently accessed content
  - Efficient content duplication cancellation across sources
- Content synchronization:
  - Real-time updates from all integrated platforms
  - Smart content grouping based on topics and timing
  - Automated source credibility scoring
  - Custom priority queuing for content processing

Summary: I like to think about this as the following; suppose there was an iPhone release. Our system simultaneously pulls information from multiple sources - news articles from NyTimes, user discussions from Reddit, tweets from X, expert reviews from tech blogs, and more. Instead of showing these as separate pieces of information, we intelligently combine them into a single, comprehensive "feed note."

For example, when Apple announces a new iPhone, our system would:

1. Detect the announcement across multiple sources
2. Group related content (news articles, social media reactions, expert analyses)
3. Remove duplicate information
4. Create a unified feed note that gives you the complete picture
5. Update this feed note in real-time as new information appears

This is all happening behind the scenes in milliseconds, ensuring you get a complete, up-to-date view of any topic without having to visit multiple websites or platforms. So when you see the “feed note” in your articles list it's all combined into one thing.

3. Machine Learning-Based Content Organization (Recommendation System)

Implementation of advanced ML algorithms for content processing:

- Custom implementation of topic modeling using LDA
- Real-time clustering using DBSCAN for content grouping
- Word embedding models for semantic similarity calculation
- Implementation of collaborative filtering for recommendations
- Custom loss function for the recommendation system
- Real-time model updating based on user interactions

Summary: The initial screen of FocusFeed is an interest screen, you might tell us you're interested in "technology" and "sports." But what does that really mean? Are you more interested in consumer tech or enterprise software? College sports or professional leagues?

Our ML system figures this out automatically by:

1. Watching how you interact with different feed notes (what you click, read, like, or skip)
2. Understanding the deeper patterns in your interests (maybe you're specifically interested in AI in healthcare, not just general tech news)
3. Finding hidden connections between topics you enjoy (perhaps articles where sports and technology intersect, like sports analytics)
4. Automatically organizing content into meaningful groups (clustering similar stories together)
5. Continuously learning and adapting as your interests evolve

For example, if you're a basketball fan who regularly reads about the NBA, our system might notice you spend extra time on articles about player statistics and team analytics. It would then start prioritizing content that combines sports with data analysis, while still maintaining a healthy mix of other basketball content. All of this happens automatically in the background, creating an increasingly personalized experience without requiring any explicit input from you beyond your normal reading behavior.

Usefulness

Personally for our team, finding reliable, relevant, and personalized content is an issue without spending hours sifting through multiple platforms, dealing with unrelated information, or encountering misinformation. FocusFeed addresses this challenge by providing a unified, intelligent platform that aggregates content from trusted news outlets, social media, and online forums into a single, coherent feed.

Users can effortlessly discover and engage with content that matters to them through an intuitive interface that learns from their interactions. The platform's real-time updates ensure users stay informed about their topics of interest, while the interactive visualization system makes exploring related content natural and engaging.

There are some existing news media sites like Google News and Flipboard but FocusFeed has its own approach to content organization and discovery and recommendation in a swiping format. Unlike existing platforms that simply collect and display news, FocusFeed creates an interconnected network of information that users can explore visually through our map as well.

The combination of recommender systems, real-time multi-source integration, and interactive visualization provides a more comprehensive and engaging way to consume news and information. The system's ability to learn from user interactions and automatically adapt its content recommendations ensures that each user's experience becomes increasingly personalized over time, rather than just what's trending right now.

Realness

Our application relies on multiple real-time APIs and datasets to provide comprehensive content coverage. Here are our primary data sources:

1. NewsAPI (newsapi.org)

- Format: JSON via REST API
- Data Volume: 30,000+ news sources
- Update Frequency: Real-time, 15-minute delay
- Content Coverage: Breaking news, articles, headlines
- Data Fields: title, description, content, publishedAt, source, url, urlToImage

- Rate Limits: 100 requests/day (free tier), 500,000 requests/month (business tier)

2. New York Times API (api.nytimes.com)

- Format: JSON via REST API
- Data Volume: 170+ years of articles
- Update Frequency: Real-time for current news
- Content Coverage: Articles, multimedia, metadata
- Data Fields: title, abstract, lead_paragraph, web_url, pub_date, section
- Rate Limits: 4,000 requests/day (free tier), 10,000 requests/day (paid tier)

3. Google News API (news.google.com/rss)

- Format: RSS/XML feed with JSON conversion
- Data Volume: Global news coverage
- Update Frequency: Real-time
- Content Coverage: News articles, headlines, topics
- Data Fields: title, link, description, pubDate, source
- Rate Limits: 100 requests/hour

4. Reddit API (reddit.com/dev/api)

- Format: JSON via REST API
- Data Volume: Access to 100,000+ subreddits
- Update Frequency: Real-time
- Content Coverage: Posts, comments, discussions, user interactions
- Data Fields: title, selftext, score, num_comments, created_utc, subreddit
- Rate Limits: 60 requests/minute per OAuth client

5. Twitter/X API (api.twitter.com)

- Format: JSON via REST API
- Data Volume: Real-time tweet stream
- Update Frequency: Real-time
- Content Coverage: Tweets, replies, user profiles, trending topics
- Data Fields: text, created_at, user, retweet_count, favorite_count

- Rate Limits: 500,000 tweets/month (Enterprise tier)

6. NOAA Climate Weather Data

- Format: JSON via REST API
- Data Volume: Extensive climate and weather records for the U.S.
- Update Frequency: Daily updates
- Content Coverage: Climate trends, extreme weather events, long-term forecasts
- Data Fields: temperature_max, temperature_min, precipitation, wind_speed
- Rate Limits: 120 requests/minute
- API Docs: https://www.ncdc.noaa.gov/cdo-web/webservice

Note: as we add more interests into the app we will have more data sources to look from, but for now this is a strong base.

Detailed User Functionality Description

A first-time visitor to FocusFeed begins by creating a personalized account through our secure authentication system. Upon their initial login, users are welcomed with an intuitive interest selection screen where they can choose from a diverse range of topics that interest them, such as technology, global affairs, sports, or cultural events. This preference selection serves as the foundation for our intelligent content curation system.

Once interests are selected, users are directed to our main recommendation interface, which presents a dynamic feed combining two content streams: personalized feed notes based on their selected interests and trending topics capturing significant global discussions. Users can interact with these feed notes through an intuitive interface – liking or disliking content, which triggers our recommendation system to continuously refine its suggestions. Under the hood, our algorithm recalibrates periodically via continuous CRUD operations on the various supported API endpoints (sources), ensuring content remains fresh and aligned with evolving user preferences.

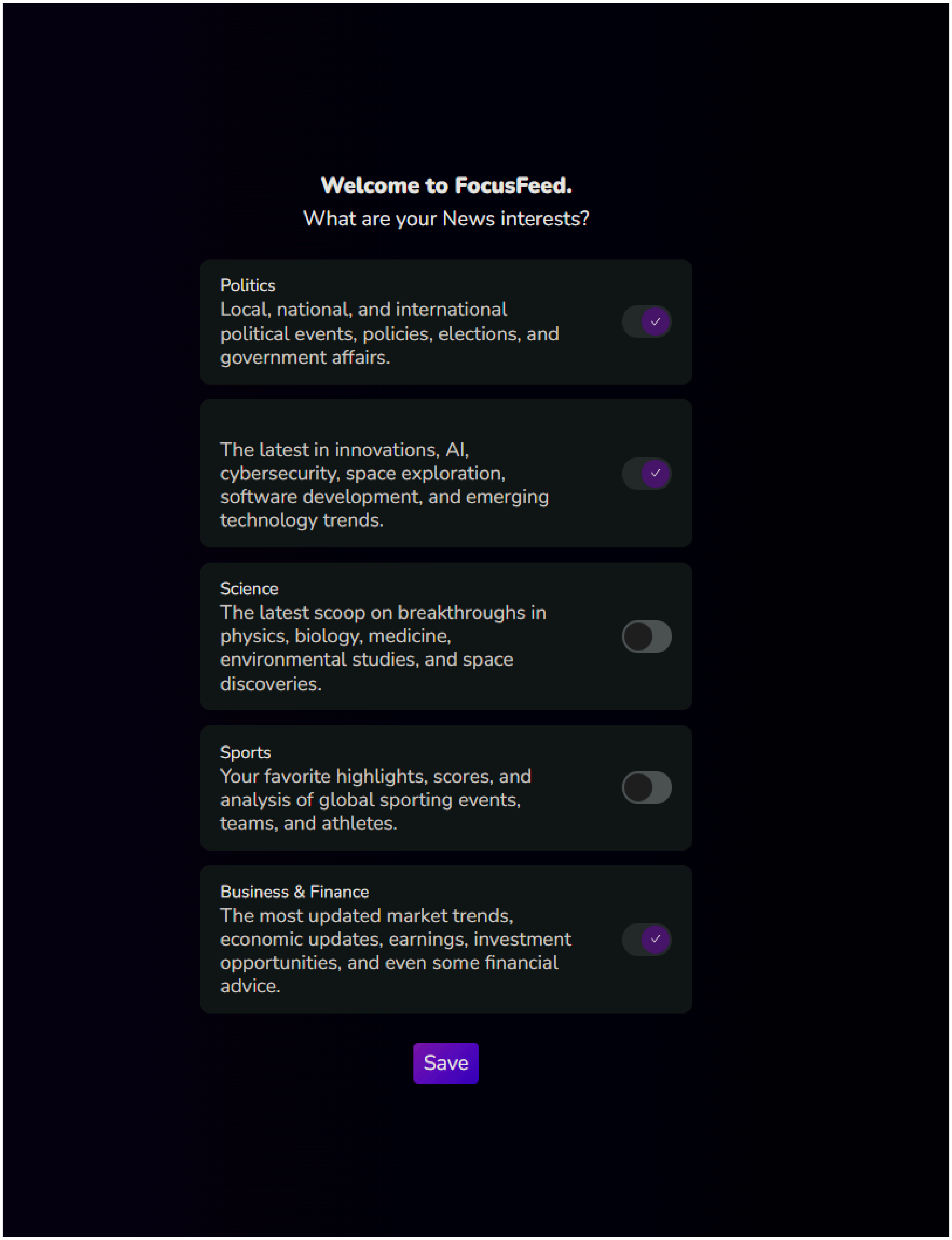
When users select a specific feed note, they enter a comprehensive view where we've seamlessly integrated information from multiple sources – news articles, social media discussions, expert analyses, and community insights – all unified into a coherent narrative. Within this view, users can engage with our intelligent RAG chatbot, which stands ready to answer specific questions about the content, provide additional context, or explore related topics in depth.

Alongside the recommendation queue, users have access to our innovative topology network map through a dedicated tab in the main interface. This custom-built visualization displays feed notes geographically, allowing users to explore content through an interactive global lens. Users can apply various filters and search to this map, adjusting their view based on topics, regions, or time periods, creating a unique and engaging way to discover relevant content across the world.

A low-fidelity UI mockup: *What do you imagine your final application's interface might look like? A PowerPoint slide or a pencil sketch on a piece of paper works!*

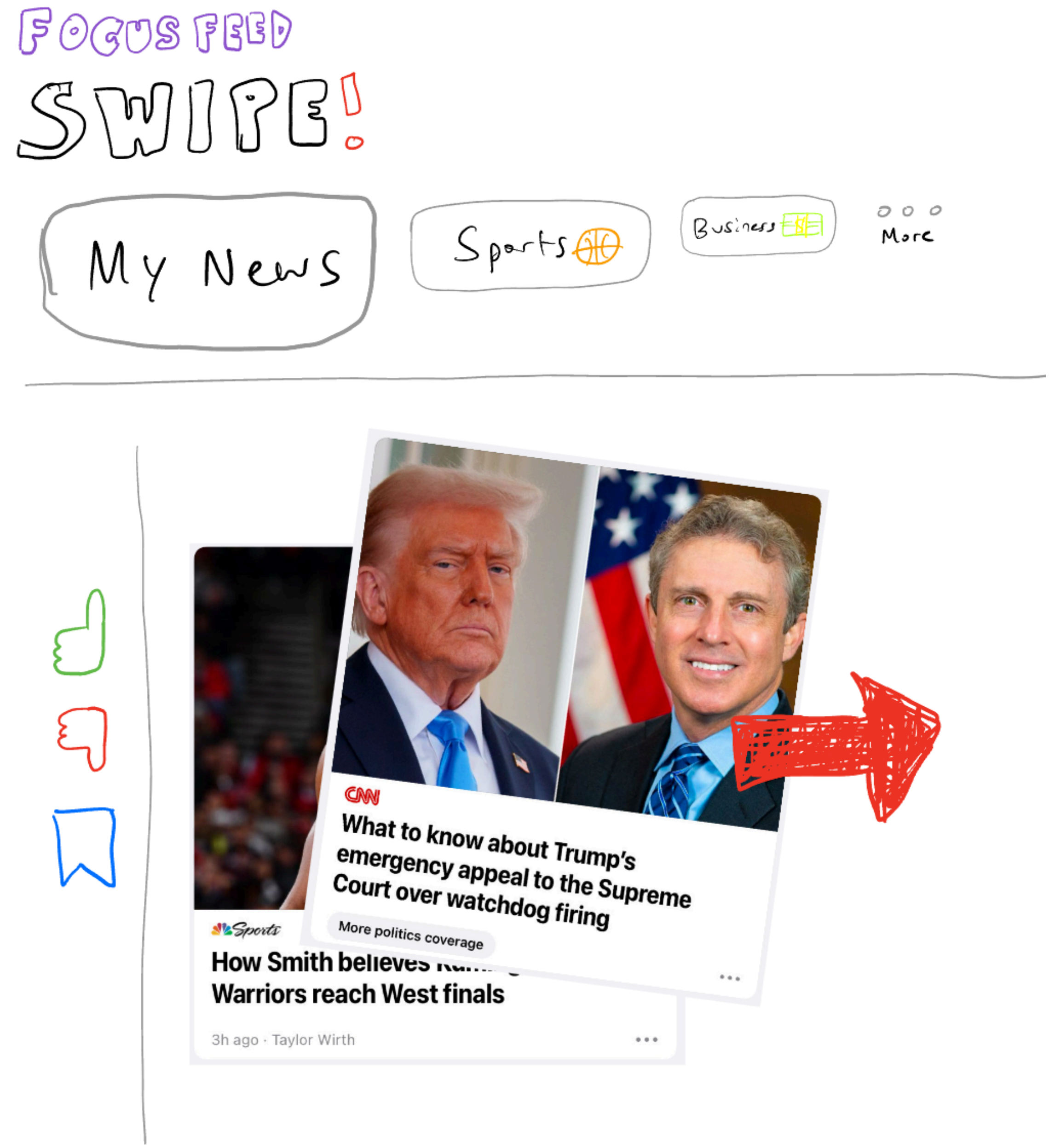
Screen #1: Initial Recommendation Selection Screen
Screen #2: Queue Rank of Article Title Screen
Screen #3: Actual Feed Notes Screen (with RAG chatbot inside it for that specific article)

Screen 1:



Welcome to FocusFeed.
What are your News interests?
Politics
Local, national, and international political events, policies, elections, and government affairs.
The latest in innovations, AI, cybersecurity, space exploration, software development, and emerging technology trends.
Science
The latest scoop on breakthroughs in physics, biology, medicine, environmental studies, and space discoveries.
Sports
Your favorite highlights, scores, and analysis of global sporting events, teams, and athletes.
Business & Finance
The most updated market trends, economic updates, earnings, investment opportunities, and even some financial advice.
Save

Screen 2:



Screen 3:

The Perfect Day In Croatia (via OpenAI)

July 14 , 2022 Comments : 35 Category : Sport

Don't Wait. The Purpose Of Our Lives Is To Be Happy!

upon arrival, your senses will be rewarded with the pleasant scent of lemongrass oil used to clean the natural wood found throughout the room, creating a relaxing atmosphere within the space.

a wonderful serenity has taken possession of my entire soul, like these sweet mornings of spring which i enjoy with my whole heart. i am alone, and feel the charm of existence in this spot, which was created for the bliss of souls like mine. i am so happy, my dear friend, so absorbed in the exquisite.

Share Marking Comment

OpenAI APIs Model used: GPT-4

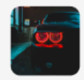
Click To See Sources

Click To Switch To Article-Only Mode

You Are Currently In Aggregate Mode. Pulling From The Following Sources:

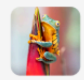
- New York Times
- Twitter (X)
- Reddit
- Google News

Recommended Posts



How To Spend The Perfect Day On Croatia's Most Magical Island

Subhead



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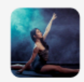

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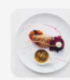

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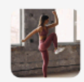

How To Spend The Perfect Day On Croatia's Most Magical Island

Subhead

Chat

FocusFeed

What Are Your Thoughts On This Article?

Me

Not Informative Enough. What Are The Latest Overall Impressions Of This Location?

FocusFeed

(Sourced From Reddit, Tripadvisor APIs) Today, Several Islan...

Type To Chat

Project work distribution

We split up the tasks and subtasks based on features so each of us took a core feature and we will build out this feature, we will all help each other with the smaller things aside from the core features.

1. Shubham More:
   a. Streaming all 3rd party API data into individual tables, ensuring we have ways to handle points of failure without bringing down the overall system.
   b. Implement the entire topological feed note map, backend + frontend
2. Anurag Chowdhury:
   a. Mainly focusing on the frontend for the topological map, will implement the auto-RAG chatbots system for every single feed note that is created.
3. Arnav Garg:
   a. Develop the initial screen that users will select what they are interested in reading about.
   b. User login and authentication screen
   c. Develop the UI for the recommendation queue page that will contain feed notes, will have all the creative flexibility for this page.
4. Michael Yan:
   a. Work on the recommendation system that is in the recommended system queue including handling the UI for that screen as well.
   b. Develops search and filtering functionality by category, integrated into the recommendation engine classification system.